Supplementary Material for GC-SE: Adversarial Example Detection in IoV via Geometric Consistency and Structural Entropy

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Appendix A
Threshold Update Strategies and Evaluation Metrics

Three threshold update strategies are compared:

Fixed Mean. The threshold is calculated using the initial benign calibration data and remains unchanged during subsequent online updates:

$$\tau_t^{\text{Fixed}} = \tau_0, \tag{1}$$

where τ_0 denotes the initial threshold.

Batch Mean. The threshold is directly updated using the current batch estimate:

$$\tau_t^{\text{Batch}} = \tau_t. \tag{2}$$

EMA Mean. The threshold is updated by combining the current batch estimate with the previous threshold through exponential moving averaging:

$$\tau_t^{\text{EMA}} = \rho\tau_{t-1}^{\text{EMA}} + (1-\rho)\tau_t, \tag{3}$$

where $\rho \in [0, 1)$ controls the contribution of the previous threshold. A larger ρ assigns greater weight to historical threshold information and reduces the influence of individual batch variations.

To isolate the effect of adversarial contamination, a clean reference is constructed for each online update round. Specifically, the injected AEs are excluded while the corresponding benign samples are retained. The threshold estimated from these benign-only samples serves as the clean reference for the corresponding round. This reference is used exclusively for offline comparison and does not participate in the actual detection or online update process.

The threshold bias is calculated as the difference between the threshold under each evaluated strategy and the corresponding clean reference threshold $\Delta\tau_t$:

$$\Delta\tau_t = \tau_t^{\text{Poisoned}} - \tau_t^{\text{Clean}}. \tag{4}$$

The recall change ΔR_t is defined as

$$\Delta R_t = R_t^{\text{Clean}} - R_t^{\text{Poisoned}}, \tag{5}$$

where positive values indicate reduced AE detection recall under the contaminated condition.

Similarly, the FPR change ΔFPR_t is calculated as

$$\Delta FPR_t = FPR_t^{\text{Poisoned}} - FPR_t^{\text{Clean}}. \tag{6}$$

Negative values indicate a reduction in FPR relative to the clean reference. All performance differences are calculated using the same evaluation samples within each round, with the corresponding thresholds applied to samples.

Appendix B
Sensitivity Analysis of Adaptive Attack Parameters

We further investigate the sensitivity of the topology-aware adaptive attack to the neighborhood size K and blending ratio α. Specifically, one parameter is varied while the other is fixed at its default value ($K = 8$ or $\alpha = 0.3$). As reported in Table X, the performance exhibits relatively small variations across different configurations. When varying K, the accuracy ranges from 98.83% to 99.17%, while the F1-score ranges from 97.91% to 98.63%. Similarly, when varying α, the accuracy ranges from 98.71% to 99.10%, and the F1-score ranges from 97.96% to 98.48%. These findings suggest that the performance is relatively stable under moderate variations in the adaptive attack parameters.

TABLE I. Sensitivity Analysis of the Topology-aware Adaptive Attack on the Shanghai Taxi + NSL-KDD Datasets

Parameter	Value	ACC	Precision	F1	AUC
α ($K = 8$)	0.10	99.02	99.39	98.32	99.22
	0.20	98.71	99.27	97.96	99.14
	0.30	**99.10**	99.41	**98.48**	99.22
	0.40	99.04	99.46	98.38	99.28
	0.50	99.00	99.49	98.36	**99.34**
K ($\alpha = 0.3$)	3	99.05	99.47	98.46	99.29
	5	**99.17**	99.41	**98.63**	**99.36**
	8	99.02	99.39	98.32	99.22
	10	99.04	99.46	98.31	99.22
	15	98.83	**99.50**	97.91	99.05

Appendix C
Sensitivity Analysis of Bandwidth

To ensure optimal performance of GC-SE, we adopt Transfer-FGSM as the representative attack for bandwidth σ

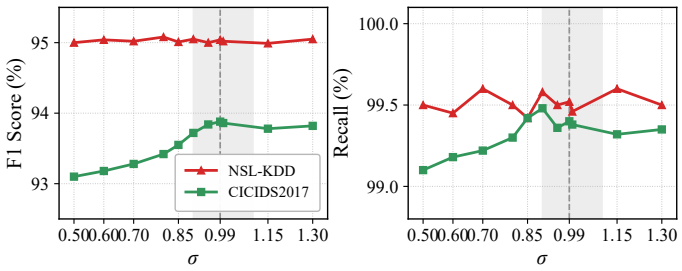


Fig. 1. Sensitivity analysis of bandwidth σ.

sensitivity analysis on the Shanghai Taxi + NSL-KDD and CICIIDS2017 datasets. This attack generates adversarial samples using a surrogate model and transfers them to the target IDS, allowing us to evaluate the robustness of graph construction against transferable adversarial perturbations without directly relying on the target model's gradients.

The bandwidth σ is varied from 0.50 to 1.30 using 11 different values, while all other experimental settings remain unchanged. As shown in Fig. **??**, the F1 score on NSL-KDD remains relatively stable around 95% across the evaluated bandwidth settings. On CICIDS2017, the F1 score gradually increases with σ and becomes relatively stable in the range of 0.90–1.30. The detection recall remains above 99% on both datasets, with only minor fluctuations. Overall, the results demonstrate that GC-SE maintains stable detection performance across a relatively broad range of bandwidth values. In particular, the default setting $\sigma = 0.99$ lies within the stable performance region. Therefore, we retain $\sigma = 0.99$ in the subsequent experiments.

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Appendix D
Implementation Details of the Joint Adaptive Attack

A. Differentiable Attack Objectives

JAA starts from adversarial samples generated by transfer-FGSM and optimizes each sample independently against the five components of GC-SE. The pretrained IDS, EAGR anchors, GCF threshold, and initial NSGA community partition are fixed during optimization.

For the IDS objective, we minimize the negative log-probability of the benign class:

$$\mathcal{L}_{\mathrm{IDS}} = -\log P_{\mathrm{IDS}}(y = 0 \mid \mathbf{x}), \tag{7}$$

where $\mathbf{x}$ denotes the candidate adversarial sample and P_{IDS} is the IDS-predicted probability.

For EAGR, we employ a soft nearest-neighbor approximation to replace discrete anchor selection. Specifically, the weights of the k nearest anchors are calculated as

$$a_j(\mathbf{x}) = \frac{\exp(-d_j(\mathbf{x})/\tau)}{\sum_{l \in \mathcal{N}_k(\mathbf{x})} \exp(-d_l(\mathbf{x})/\tau)}, \tag{8}$$

where $d_j(\mathbf{x}) = \|\mathbf{x} - \mathbf{u}_j\|_2^2$ is the squared Euclidean distance to anchor $\mathbf{u}_j$, $\mathcal{N}_k(\mathbf{x})$ denotes the selected anchor neighborhood, and τ is the temperature parameter. The weighted anchor-label aggregation provides the estimated benign-class probability $P_{\mathrm{EAGR}}(y = 0 \mid \mathbf{x})$, yielding

$$\mathcal{L}_{\mathrm{EAGR}} = -\log P_{\mathrm{EAGR}}(y = 0 \mid \mathbf{x}). \tag{9}$$

For GCF, a differentiable geometric inconsistency surrogate is constructed from the IDS and EAGR probability vectors:

$$\begin{aligned}\widetilde{D}_{\mathrm{GCF}}(\mathbf{x}) = \|\mathbf{p}_{\mathrm{IDS}}(\mathbf{x})\|_2 + \|\mathbf{p}_{\mathrm{EAGR}}(\mathbf{x})\|_2 \\ - \|\mathbf{p}_{\mathrm{IDS}}(\mathbf{x}) + \mathbf{p}_{\mathrm{EAGR}}(\mathbf{x})\|_2.\end{aligned} \tag{10}$$

The corresponding objective penalizes scores exceeding the GCF threshold θ_{GCF}:

$$\mathcal{L}_{\mathrm{GCF}} = \max(0, \widetilde{D}_{\mathrm{GCF}}(\mathbf{x}) - \theta_{\mathrm{GCF}}). \tag{11}$$

B. Topology and Structural-Entropy Surrogates

Since the original graph construction and NSGA-based community optimization are non-differentiable, JAA employs feature-based approximations to incorporate community-related information into adversarial optimization.

Specifically, nodes from benign-majority communities and other communities in the initial partition are collected as two reference sets, denoted by $\mathcal{R}_b$ and $\mathcal{R}_o$, respectively. Their feature affinities to a candidate sample are estimated using an RBF kernel:

$$K(\mathbf{x}, \mathbf{z}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{z}\|_2^2}{2h^2}\right), \tag{12}$$

where $\mathbf{z}$ is a reference feature vector and h is the kernel bandwidth.

The topology-related objective encourages the candidate to approach benign-majority communities while reducing its affinity to other communities:

$$\mathcal{L}_{\mathrm{topology}} = \lambda_{\mathrm{in}}(1 - S_b(\mathbf{x})) + \lambda_{\mathrm{out}} S_o(\mathbf{x}), \tag{13}$$

where S_b and S_o denote the average RBF affinities to $\mathcal{R}_b$ and $\mathcal{R}_o$, respectively, and λ_{in} and λ_{out} control their contributions.

For the structural-entropy component, we use the average Euclidean distance to the five nearest nodes in the benign-majority reference set as a surrogate:

$$\widetilde{\mathcal{L}}_{\mathrm{SE}} = \frac{1}{5} \sum_{\mathbf{z} \in \mathcal{N}_5^b(\mathbf{x})} \|\mathbf{x} - \mathbf{z}\|_2, \tag{14}$$

where $\mathcal{N}_5^b(\mathbf{x})$ denotes the five nearest reference nodes in $\mathcal{R}_b$.

C. Joint Optimization and Exact Validation

The five objectives are combined using the weighted formulation. The candidate features are optimized using Adam for 250 iterations. After optimization, all generated samples are evaluated using the original GC-SE pipeline. For each candidate, the original graph and community structure are independently copied from the initial deployment state to avoid interference between evaluations. The final detection decision follows the original GC-SE rule.